Emilio Bardán Negrete

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Research interests: mechanistic interpretability; geometric deep learning and Lie group representations; protein structure models (AlphaFold, RoseTTAFold); applied algebraic topology.

EDUCATION

Instituto Tecnológico Autónomo de México (ITAM) Aug 2023 – Jun 2027 (expected)
B.S. in Applied Mathematics Mexico City

- GPA 8.69 / 10 (36 courses; minimum passing grade 6)
- *Relevant coursework:* Linear Algebra I–II; Mathematical Analysis I–II; Abstract Algebra; Probability I–II; Complex Variable; Numerical Analysis; Dynamical Systems I–II; Computational Mathematics; Automata Theory; *In progress (Fall 2026)*: Numerical Optimization I; Partial Differential Equations; Mathematical Statistics; Stochastic Processes.

Singapore Management University Summer 2026
Summer exchange program Singapore

- *Relevant coursework:* Quantum Computing for Financial Services.

RESEARCH

Interpreting AlphaFold's Pairformer October 2026 – present
Bachelor's thesis, ITAM · Advisor: Mario Vázquez Corte

- Investigates, via activation patching, whether the network encodes intrinsic residue properties (e.g. hydrophobicity) that it translates into geometry, and how evolutionary signals propagate into its predictions.

The geometry of a mental task: Topological Data Analysis for EEG processing Oct 2026
E. Bardán. Preprint, Neurolab, ITAM

- Built a pipeline that computes persistent homology of EEG functional connectivity; an SVM on the resulting features classifies rest vs mental arithmetic (36 subjects) with 89% accuracy under nested, subject-wise cross-validation, on par with band-power baselines. The paper also addresses a common gap in TDA+EEG literature: it provides a self-contained development of the algebraic topology underlying the method.

Information decay on the Lorenz attractor May 2026
Course research project

- Estimated mutual-information decay with the KSG estimator across sweeps of ρ; introduced a burstiness measure on the derivative of $I(t)$.

ACADEMIC SERVICE

President, Applied Mathematics Student Board Jan – Dec 2026
ITAM

- Advocated for ITAM's first participation in the Putnam Competition (Dec 2026) and helped build the preparation program.
- Proposed the new course "Philosophy and Mathematics" and secured faculty approval; first offered Spring 2027.
- Organized and moderated two panels: on consciousness and the viability of AGI (Mar 2026), and on whether machines can do mathematics (Nov 2026).
- Organized 4 guest talks on AI, mathematics, and physics, with speakers including David Fernández-Duque (University of Barcelona), Darío Núñez (Institute of Nuclear Sciences, UNAM), and Pablo Montero (Google Cloud México).
- Ran mathematics community events (Integration Bee, movie nights, student–faculty luncheons).

WRITING AND OUTREACH

- "El reto de la inteligencia artificial." *El Heraldo de México*, no. 3065, p. 19, Nov 6, 2025. Opinion column.
- "El universo como información." *Laberintos & Infinitos*, forthcoming Nov 2026.
- *El Supuesto* (ITAM student newspaper): 9 science and technology articles, 2024–2026, on Gödel, quantum computing, and AI.
- Essays on intelligence and consciousness, including "The thinking evolution," at emiliobardan.substack.com.

CERTIFICATES

- **MIT xPRO:** Deep Learning: Mastering Neural Networks (Aug 2024); Quantum Computing Fundamentals (Apr 2025); Machine Learning, Modeling, and Simulation: Engineering Problem-Solving in the Age of AI (Jul 2026).
- **IBM / Coursera:** Machine Learning with Python; Introduction to Deep Learning and Neural Networks with Keras; Deep Learning with Keras and TensorFlow (2025).

SKILLS

Programming	Python (PyTorch, TensorFlow, scikit-learn, NumPy, SciPy), MATLAB, LaTeX, Jupyter
Languages	Spanish (native); English (C1, TOEFL iBT 110/120); French (B1, DELF B1; Florida Seal of Biliteracy, Silver)